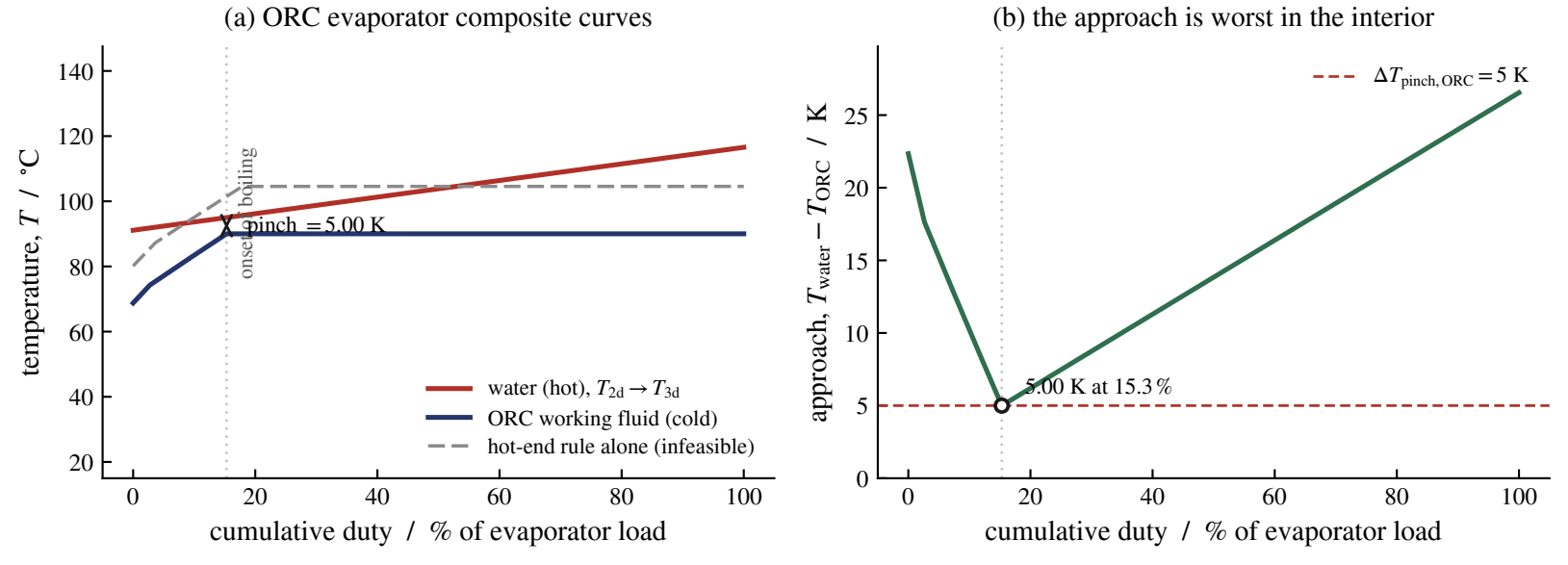reference case (R4)